AREC 310 - Homework 3 - Answer Key

30 points possible

Directions: Answer the following questions completely. You can work with your peers, but you MUST TURN IN YOUR OWN ANSWERS (i.e., in your own words) to get credit. Upload a PDF to Canvas by the due date.

Problem 4: Impact of Higher Beef Prices on the Pork Market (11 points)

Suppose you are a pork market analyst, and beef prices are expected to rise dramatically, by 10%.

Because beef and pork are substitutes, this price change creates a demand shock in the pork market. Specifically, you are asked to calculate the resulting changes in the pork price and pork quantity.

Parameters:

- Long-run supply elasticity of pork: $E_S = 2.15$
- Long-run demand elasticity of pork: $E_D = -1.96$
- Cross-price elasticity of pork demand with respect to beef: 0.6

Tasks:

1. Calculate the value of the demand shock, S_D.
2. Write the supply and demand equations in elasticity form.
3. Solve for the equilibrium percent change in pork price, $\%\Delta P$.
4. Solve for the equilibrium percent change in pork quantity, $\%\Delta Q$.
5. Briefly explain the intuition behind your results.

Hint: Recall that $S_D = (\textit{cross-price elasticity})(\%\Delta P_{beef})$

Answer Key

Step 1. Demand shock

$$S_D = (0.6)(10) = 6\%$$

Because the shock comes from beef prices (a substitute), it shifts the pork demand curve to the right.
Supply is unaffected, so $S_S = 0$.

Step 2. Write supply and demand equations

Supply:

$$\%\Delta Q_S = E_S(\%\Delta P) + S_S = 2.15(\%\Delta P) + 0$$

Demand:

$$\%\Delta Q_D = E_D(\%\Delta P) + S_D = -1.96(\%\Delta P) + 6$$

Step 3. Impose equilibrium and solve for $\%\Delta P$

$$\begin{aligned}
\%\Delta Q_S &= \%\Delta Q_D \\
2.15(\%\Delta P) &= -1.96(\%\Delta P) + 6 \\
(2.15 + 1.96)(\%\Delta P) &= 6 \\
\%\Delta P &= \frac{6}{4.11} \\
\%\Delta P &\approx 1.46\%
\end{aligned}$$

So pork prices are expected to rise **1.46%**.

Step 4. Solve for $\%\Delta Q$

Supply side:

$$\%\Delta Q_S = 2.15(1.46) = 3.14\%$$

Demand side:

$$\%\Delta Q_D = -1.96(1.46) + 6 = -2.86 + 6 = 3.14\%$$

Both agree: quantity rises by **3.14%**.

Step 5. Intuition

- The 10% increase in beef prices makes pork more attractive.
- Demand shifts rightward ($S_D = +6\%$).
- Pork prices rise modestly (+1.46%), while pork output increases more strongly (+3.14%).
- The results are consistent with elasticities: because supply is relatively elastic in the long run, much of the adjustment occurs through higher output rather than sharply higher prices.